

Classical Models — Sample Output, Midterm Scale vs. Full Dataset

NLP Final Term Project, Group 02. Naive Bayes, Logistic Regression and Support Vector Machine, bag-of-words and TF-IDF, both datasets. Same layout the supervisor's sample output specifies, at two points in the project.

1. Midterm scale. Balanced to 3,000 documents per class per dataset (6,000 rows total per dataset), split 4,320 train / 480 validation / 1,200 test with `random_state=42`, stratified. These are the numbers submitted at midterm, reproduced here exactly from the saved reproduction record in `experiments/midterm/` and re-verified against the originally submitted values before this document was built.

Sample output:

Naive Bayes

Representation	Dataset	Accuracy	Precision	Recall	F1
BoW	Dataset 1	0.9567	0.9580	0.9567	0.9566
TF-IDF	Dataset 1	0.9508	0.9515	0.9508	0.9508
BoW	Dataset 2	0.8583	0.8587	0.8583	0.8583
TF-IDF	Dataset 2	0.8275	0.8387	0.8275	0.8261

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Logistic Regression

Representation	Dataset	Accuracy	Precision	Recall	F1
BoW	Dataset 1	0.9825	0.9826	0.9825	0.9825
TF-IDF	Dataset 1	0.9758	0.9761	0.9758	0.9758
BoW	Dataset 2	0.9333	0.9346	0.9333	0.9333
TF-IDF	Dataset 2	0.8925	0.8926	0.8925	0.8925

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Support Vector Machine

Representation	Dataset	Accuracy	Precision	Recall	F1
BoW	Dataset 1	0.9775	0.9776	0.9775	0.9775
TF-IDF	Dataset 1	0.9875	0.9877	0.9875	0.9875
BoW	Dataset 2	0.9117	0.9132	0.9117	0.9116
TF-IDF	Dataset 2	0.9083	0.9084	0.9083	0.9083

2. Full dataset, final term. The full balanced corpora, no sampling: 34,994 rows for DAIGT V2, 53,806 rows for HC3, the same group-aware 72/8/20 split every model in the final report uses. Recomputed from the saved per-document scores in `experiments/audit/` and cross-checked against the recorded accuracy and F1 before this document was built.

Sample output:

Naive Bayes

Representation	Dataset	Accuracy	Precision	Recall	F1
BoW	Dataset 1	0.9591	0.9603	0.9591	0.9591
TF-IDF	Dataset 1	0.9577	0.9578	0.9577	0.9577
BoW	Dataset 2	0.8713	0.8715	0.8713	0.8713
TF-IDF	Dataset 2	0.8672	0.8698	0.8672	0.8670

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Logistic Regression

Representation	Dataset	Accuracy	Precision	Recall	F1
BoW	Dataset 1	0.9893	0.9893	0.9893	0.9893
TF-IDF	Dataset 1	0.9857	0.9858	0.9857	0.9857
BoW	Dataset 2	0.9551	0.9551	0.9551	0.9551
TF-IDF	Dataset 2	0.9365	0.9365	0.9365	0.9365

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Support Vector Machine

Representation	Dataset	Accuracy	Precision	Recall	F1
BoW	Dataset 1	0.9870	0.9870	0.9870	0.9870
TF-IDF	Dataset 1	0.9910	0.9911	0.9910	0.9910
BoW	Dataset 2	0.9475	0.9476	0.9475	0.9475
TF-IDF	Dataset 2	0.9449	0.9452	0.9449	0.9449

What changed between the two. More data, not a different method: same three models, same two representations, same code path. Naive Bayes on HC3 moves from 0.8583 to 0.8713 F1 (BoW) as the extra rows help it estimate word frequencies more reliably, the model in the report Naive Bayes is weakest at is the one it is expected to be weakest at, an independence assumption text does not satisfy. Logistic Regression and the SVM move by less, because a smaller sample already gave their decision boundary enough to work with. No representation switches which one wins between the two scales, BoW stays ahead of TF-IDF for Naive Bayes and Logistic Regression on both datasets, TF-IDF stays ahead for the SVM on DAIGT V2.